



Gazi Üniversitesi — Bilgisayar Muhendisliği Bölümü

CENG-479 Parallel Programming

Submission 1 — Project Proposal

Spring 2026 | Deadline: 17 May 2026

Course	CENG-479 Parallel Programming
Language	English
Team Size	2 students (mandatory)
Technologies	Java Threads POSIX Threads CUDA
Submission	Submission 1 of 2
Deadline	17 May 2026
Instructor	Asst. Prof. Dr. Huseyin Temucin

Objective

Each 2-student team will propose a parallel algorithm-based solution for a computationally intensive real-world problem. The proposal must identify a suitable problem, select a parallelization strategy, and justify the approach with at least 3 peer-reviewed academic references.

Allowed Technologies (select one)

- **Java Threads — Multi-threading using Thread, Runnable, ExecutorService, ForkJoinPool, etc.**
- **POSIX Threads (pthreads) — C/C++ shared-memory parallelism on Linux/Unix.**
- **CUDA — GPU-based parallel computing using NVIDIA CUDA.**

Project Requirements

- **The problem must genuinely benefit from parallelization (sorting, matrix operations, image processing, simulations, graph algorithms, etc.).**
- Include a theoretical analysis of why the problem is parallelizable.
- **Cite at least 3 peer-reviewed academic references (journals / conference proceedings).**
- Outline the expected speedup or performance improvement.
- Plan a comparison between sequential and parallel performance.

Required Document Sections

1. Project Title and Team Members
2. Problem Statement — What computationally intensive problem will be solved?

3. Motivation — Why does this problem benefit from parallelism?
4. Proposed Parallel Algorithm — Description and strategy
5. Selected Technology — Java Threads / POSIX Threads / CUDA (with justification)
6. Performance Analysis Plan — How will you measure speedup?
7. Academic References — Minimum 3 peer-reviewed sources (APA format)

Grading Criteria

Criterion	Weight	Points
Suitability of Problem for Parallelism	25%	25
Quality of Academic References (min. 3)	25%	25
Clarity of Algorithm Description	20%	20
Justification of Technology Choice	15%	15
Performance Analysis Plan	15%	15

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17 May 2026

Submit via Google Classroom. The proposal must cite at least 3 peer-reviewed academic sources.